

Multi-Stage Robust Optimization of a Wastewater Treatment Biogas Generator Participating in Day-Ahead and Real-Time Regulation Markets

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Abstract

Wastewater treatment plants are large, energy-intensive loads that can be optimally controlled to support grid reliability and reduce operational costs. This paper presents a multi-stage robust optimization framework to control a wastewater treatment plant equipped with a biogas generator and storage tank for participation in California's day-ahead and real-time frequency regulation markets while managing biogas production and regulation signal uncertainty. We solve for day-ahead regulation capacity in the first stage and real-time regulation capacity over the day as the uncertainty is progressively revealed. We propose affine control policies to determine the real-time regulation capacity based on partial uncertainty realizations. This results in a tractable affinely adjustable robust counterpart. In a case study, we evaluate our proposed approach against a day-ahead-only robust formulation and found that our approach increases regulation capacity provision and lowers operational costs.

Keywords: biogas, demand response, regulation, adjustable robust optimization, wastewater treatment plants

1. Introduction

After years of relatively stagnant growth, the U.S. Energy Information Administration recently forecasted an increase in national electricity consumption in 2025 and 2026, with much of this growth stemming from data centers and industrial loads [1]. Increasing operational flexibility—such as through demand response—may play an important role in helping grid operators meet growing demand. Demand response is a mechanism for consumers to reduce or shift their energy consumption in response to financial benefits, usually to aid operators during critical periods of peak load.

Industrial and utility-scale loads are particularly well-suited to participate in demand response. These

consumers are typically large electricity users equipped with the communication and control infrastructure necessary for participating in demand response. Additionally, electricity costs are often large portions of their operating costs, creating strong financial incentives to participate in such programs [2, 3]. Demand response has been studied in a range of industrial processes and utilities, including cement crushing plants [4], steel plants [5], aluminum smelting [6], data centers [7], and drinking water networks [8]. In this work, we specifically focus on wastewater treatment plants (WWTPs) as flexible loads that can simultaneously provide grid services and reduce operation costs.

WWTPs share many of the same characteristics as other promising demand response resources. First, wastewater treatment is an energy-intensive process. For many municipalities, WWTPs are the largest single energy consumer [9], and nationally they account for 1% of the total electricity consumption (i.e., 30.2 TWh/year) [10]. Due to their large, centralized nature, WWTPs may offer more immediate and reliable flexibility than aggregations of smaller distributed loads. Second, most modern WWTPs have supervisory control and data acquisition (SCADA) systems, which enable real-time monitoring of system parameters (e.g., flow rates, pH levels, and tank levels) and automation of plant equipment (e.g., pumps, valves, and variable frequency drives) [11, 12]. This infrastructure makes it technically feasible to adjust operation in response to external signals. Third, WWTPs are also incentivized to participate in demand response programs. Electricity costs can be 25% to 40% of a WWTP's annual operating budget [13]. The Environmental Protection Agency (EPA) estimates that municipal WWTPs across the United States spend a combined \$2 billion in annual electric costs [14]. Furthermore, WWTPs can have on-site infrastructure to delay treatment processes (and hence, delay the plant's energy consumption) via influent storage tanks, as well as offset their energy consumption from the grid via biogas generators [15]. Given these characteristics, WWTPs are good candidates to provide grid services

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through demand response. Increased coordination between the power and wastewater sectors could reduce operational costs for both these sectors, as well as increase reliability for the power grid.

In this paper, we focus on a WWTP with an on-site biogas generator and biogas storage tank participating in the day-ahead and real-time frequency regulation markets. Biogas is produced during the WWTP’s anaerobic digestion stage and can be used to generate electricity. The generated electricity from biogas can significantly offset the electricity needs of a WWTP: combined heat and power systems fueled by recovered biogas can meet almost 50% to 70% of a WWTP’s electricity needs [16]. Through strategic use of the on-site biogas generator and storage tank (i.e., reducing or shifting the net power demand), plant operators can maximize their profits from participating in the day-ahead and real-time frequency regulation markets.

Several studies have investigated the feasibility of WWTP demand response participation. In [17], the authors use a digital twin to simulate energy flexibility resources at a WWTP, including on-site batteries, biogas storage, and influent storage. In [18] and [19], WWTP operation is adjusted to reduce electricity costs, where [18] modeled load shifting of decentralized wastewater treatment, considering peak and off-peak electric rates through simulations and [19] dispatched biogas generators to minimize electricity costs. However, none of these works considered uncertainty or explicitly modeled ancillary service participation with the exception of [20]. Anaerobic digestion is a complex process, and the challenge of accurately predicting biogas production has motivated recent work on machine learning models to improve methane yield prediction [21]. Ref. [20] optimally solves for the regulation capacities in CAISO’s day-ahead market while managing biogas production uncertainty. While effective, the robust formulation can be conservative: as the uncertainty set increases, the WWTP stores more biogas in the storage tank to make sure the biogas volume lower limit is not violated under all possible uncertainty realizations. Additionally, the WWTP operator flares excess biogas to avoid violating the biogas volume upper limit. Because of this, less biogas is used to offset the plant’s power demand and less regulation capacity is bid into the ancillary services market.

To address this, we propose a multi-stage robust optimization framework that solves for the WWTP’s participation in the day-ahead and real-time frequency regulation market. Using affine control policies to determine the real-time regulation capacity based on the biogas and regulation signal uncertainty, we solve this biogas scheduling and regulation capacity problem as an affinely adjustable robust problem. Participating in

the real-time market could help reduce flaring, increase the regulation capacity that a WWTP can provide, and decrease WWTP operational costs by being able to provide additional regulation capacity once uncertainty becomes known over the day.

The main contributions of this paper are i) developing a multistage robust formulation that solves for biogas generator dispatch and day-ahead and real-time regulation market participation subject to biogas production and regulation signal uncertainty, ii) employing affine control policies that solve for time-delayed real-time regulation capacity decisions based on partial uncertainty realizations, iii) tractably reformulating the multi-stage robust approach given the affine control policy, and iv) comparing our approach against a purely day-ahead robust approach in a detailed case study to assess performance trade-offs.

The rest of the paper is organized as follows. Section 2 presents the formulation for the WWTP biogas operation, regulation participation, and deterministic formulation. In Section 3, we develop the multistage robust formulation. Section 4 evaluates the performance of the approach in a case study and Section 5 provides concluding remarks.

2. Problem Description

In this work, we develop an optimization framework to schedule and control the operation of an on-site biogas generator and biogas storage tank at a WWTP. The generator enables the WWTP to participate in both day-ahead and real-time frequency regulation services. We solve the problem over a 24-hour horizon, accounting for the different timescales and market structures associated with day-ahead and real-time ancillary services. In our formulation, the day-ahead regulation capacity is solved over one-hour blocks (i.e., $\Delta T_{DA} = 1$ hour) and the real-time regulation capacity is solved over 15-minute intervals (i.e., $\Delta T_{RT} = 0.25$ hour). We use subscript h to denote the trading hour and subscript i to denote the 15-minute interval within the trading hour. The full set of 15-minute intervals within a day is represented as $(h, i) \in \mathcal{T}$, while all one-hour periods are represented as $h \in \mathcal{T}_h$.

The remainder of this section introduces the key components of the formulation: the WWTP biogas model, the frequency regulation capacity model, the associated costs, and the deterministic version of the problem that assumes perfect knowledge of the biogas production and regulation signal. In Section 3, we extend this formulation to incorporate biogas and regulation signal uncertainty, resulting in a multi-stage robust formulation which we solve using affine control policies.

2.1. WWTP Biogas Operation

We model and optimally control an on-site biogas generator at the WWTP. Biogas is produced as a by-product of anaerobic digestion, which is the solids management stage of the WWTP plant for treating sewage sludge (see [20, 22] for more details). This biogas can be used to offset the plant's electricity demand, where $p_g^{h,i}$ denotes the net power demand of the WWTP (i.e., demand minus generation) at time (h, i)

$$p_g^{h,i} = p_{\text{demand}}^{h,i} - p_b^{h,i} \quad \forall (h, i) \in \mathcal{T}, \quad (1)$$

where $p_{\text{demand}}^{h,i}$ is the plant's base power demand and $p_b^{h,i}$ is the power generated from biogas. The biogas generator's power output $p_b^{h,i}$ is proportional to the amount of biogas consumed $b_{\text{out}}^{h,i}$ at time (h, i)

$$p_b^{h,i} = \frac{\psi}{\Delta T_{\text{RT}}} \cdot b_{\text{out}}^{h,i} \quad \forall (h, i) \in \mathcal{T}, \quad (2)$$

where parameter ψ converts the volume of biogas consumed (in cubic meters) into electric energy produced over period (h, i) of duration ΔT_{RT} . The generator's power output must be within its limits when it is on and forced to zero when the generator is off

$$\underline{p}_b \cdot z^h \leq p_b^{h,i} \leq \overline{p}_b \cdot z^h \quad \forall (h, i) \in \mathcal{T}, \quad (3)$$

$$b_{\text{out}}^{h,i} \geq 0 \quad \forall (h, i) \in \mathcal{T}, \quad (4)$$

$$z^h \in \{0, 1\} \quad \forall h \in \mathcal{T}_h, \quad (5)$$

where binary decision variable z^h is the on/off status of the biogas generator within hour h , and parameters $\overline{p}_b$ and $\underline{p}_b$ are the maximum and minimum power output of the generator when it is on. We ensure that the biogas consumed is non-negative in (4). To prevent wear from the biogas generator switching on and off frequently, we limit the biogas generator status to be constant over each trading hour.

The generator's operation is further constrained by the amount of biogas available, which is governed by volume of biogas stored in the storage tank

$$v^{h,i} = v^{(h,i)-1} + b_{\text{in}}^{h,i} - b_{\text{out}}^{h,i} - q^{h,i} \quad \forall (h, i) \in \mathcal{T}, \quad (6)$$

$$q^{h,i} \geq 0 \quad \forall (h, i) \in \mathcal{T}, \quad (7)$$

$$0 \leq v^{h,i} \leq \bar{v} \quad \forall (h, i) \in \mathcal{T}, \quad (8)$$

where $v^{h,i}$ is the volume of biogas available at time (h, i) , $b_{\text{in}}^{h,i}$ is the amount of biogas produced from anaerobic digestion at time (h, i) , $q^{h,i}$ is the amount of biogas flared, and $\bar{v}$ is the maximum volume of

biogas that can be stored in the tank. In (6), the available volume of biogas depends on the volume from the previous time period, the amount of biogas being produced, and the amount of biogas being used by the generator or flared. Flaring is common at WWTPs, where less than 10% of biogas produced at WWTPs currently goes toward a productive use, such as generating electricity and/or heat [23]. While we can produce electricity from biogas, there is an added cost associated with processing the biogas to remove contaminants that could damage the generator (i.e., scrubbing costs). These processing costs are incorporated into the objective function in Section 2.3. In our formulation, we treat the biogas produced as uncertain; this is discussed further in Section 3.

2.2. Day-Ahead and Real-Time Regulation

We model the WWTP's participation in both the day-ahead and real-time ancillary services market for frequency regulation services. Since our case study in Section 4 uses data about a WWTP located in California, we frame our description and formulation of regulation services given the market structure of the California Independent System Operator (CAISO) [24].

In CAISO, participants submit bids for their capability to adjust their power consumption or generation around their baseline in real time. Once awarded, the participants must follow an automatic control signal (ACS) sent by the ISO. This signal is updated every four seconds (i.e., $\Delta T_{\text{reg}} = 1/900$ hours), and instructs participants to increase or decrease their power generation or consumption away from their baseline in proportion to their awarded regulation capacity.

Regulation capacity is procured at different timescales. Day-ahead regulation awards are made in one-hour blocks, while real-time regulation is awarded in 15-minute intervals. The day-ahead market closes at 10 am the day before where results are posted by 1 pm [24]. In contrast, the real-time ancillary services market closes 75 minutes before the trading hour and the results are published 45 minutes before [24]. These timelines are illustrated in Fig. 1.

CAISO allows for asymmetric regulation participation, with separate capacity bids for regulation up (i.e., increase power generation or decrease power demand) and regulation down (i.e., decrease power generation or increase power demand). The regulation signal $\tilde{s}$ can be split into its positive and negative components, corresponding to regulation up and down, respectively (see Fig. 2). The real-time power demand setpoint at each four-second period ℓ within time step

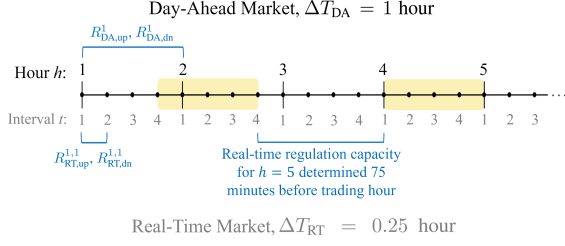


Figure 1. CAISO regulation bidding timeline.

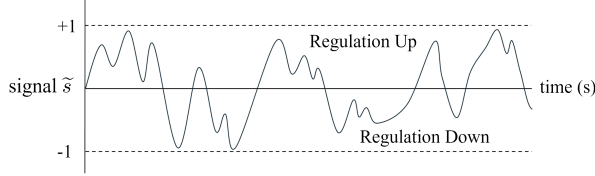


Figure 2. Example signal where positive values are regulation up (increase generation) and negative values are regulation down (decrease generation).

(h, i) can be expressed as

$$p^{h,i,\ell} = p_0^{h,i} + \begin{cases} (R_{DA,dn}^h + R_{RT,dn}^{h,i}) \tilde{s}^{h,i,\ell} & \text{if } \tilde{s}^{h,i,\ell} < 0 \\ (R_{DA,up}^h + R_{RT,up}^{h,i}) \tilde{s}^{h,i,\ell} & \text{if } \tilde{s}^{h,i,\ell} \geq 0 \end{cases}$$

where $p^{h,i,\ell}$ is the real-time power setpoint, $p_0^{h,i}$ is the scheduled (or baseline) power setpoint, $R_{DA,up}^h$ and $R_{DA,dn}^h$ are the day-ahead regulation up and down capacities, and $R_{RT,up}^{h,i}$ and $R_{RT,dn}^{h,i}$ are the real-time regulation up and down capacities. These capacities must be non-negative

$$R_{DA,up}^h, R_{DA,dn}^h \geq 0 \quad \forall h \in \mathcal{T}_h, \quad (9)$$

$$R_{RT,up}^{h,i}, R_{RT,dn}^{h,i} \geq 0 \quad \forall (h, i) \in \mathcal{T}. \quad (10)$$

We also need to ensure that the generator's power output remains within its limits when providing regulation, so (3) is updated as

$$p_b^{h,i} + R_{DA,up}^h + R_{RT,up}^{h,i} \leq \bar{p}_b \cdot z^h \quad \forall (h, i) \in \mathcal{T}, \quad (11)$$

$$p_b^{h,i} - R_{DA,dn}^h - R_{RT,dn}^{h,i} \geq \underline{p}_b \cdot z^h \quad \forall (h, i) \in \mathcal{T}. \quad (12)$$

Last, we ensure that there is sufficient biogas available in the storage tank to provide regulation

$$0 \leq v^{h,i} - \sum_{\ell} \left(\frac{\Delta T_{reg}}{\psi} \cdot (R_{DA,dn}^h + R_{RT,dn}^{h,i}) \cdot \tilde{s}_{dn}^{h,i,\ell} + \frac{\Delta T_{reg}}{\psi} \cdot (R_{DA,up}^h + R_{RT,up}^{h,i}) \cdot \tilde{s}_{up}^{h,i,\ell} \right) \leq \bar{v}, \quad (13)$$

for all $(h, i) \in \mathcal{T}$, where $\tilde{s}_{up}^{h,i,\ell} \triangleq \max(\tilde{s}^{h,i,\ell}, 0)$ and $\tilde{s}_{dn}^{h,i,\ell} \triangleq \min(\tilde{s}^{h,i,\ell}, 0)$. For symmetric regulation

services, an additional constraint can be incorporated to enforce that regulation up capacity is equal to regulation down capacity, i.e.,

$$R_{DA}^h \triangleq R_{DA,up}^h = R_{DA,dn}^h \quad \forall h \in \mathcal{T}_h, \quad (14)$$

$$R_{RT}^{h,i} \triangleq R_{RT,up}^{h,i} = R_{RT,dn}^{h,i} \quad \forall (h, i) \in \mathcal{T}. \quad (15)$$

Symmetric regulation may be a requirement of the program (e.g., PJM's RegD service) or may be a choice of the regulation participation (e.g., an energy-neutral battery) [25]. In our robust formulation in Section 3, we focus on symmetric regulation to aid in tractably reformulating the problem.

2.3. Objective Function

To capture the costs associated with biogas generator control and frequency regulation participation, the objective function includes both scheduled operation costs as well as revenue from providing day-ahead and real-time regulation services.

2.3.1. Scheduled Costs. The scheduled cost includes the cost of electricity and biogas processing as well as tank volume and flaring penalties. For large consumers—such as WWTPs—the electricity cost typically has both an energy charge and demand charge. The energy charge is based on the amount of energy consumed, while the demand charge is calculated based on the maximum power demand of the consumer over the entire billing period. There may also be separate demand charges for certain periods of the day, i.e., time-of-use (TOU) rates. We denote p_D^k as the maximum power demand during TOU period k

$$p_D^k \geq p_g^{h,i} \quad \forall (h, i) \in \mathcal{T}_k, k \in \text{TOU}, \quad (16)$$

$$p_D^k \geq p_D^{k,0} \quad \forall k \in \text{TOU}, \quad (17)$$

where parameter $p_D^{k,0}$ is the maximum power demand for TOU period k from previous days within the billing period and $\mathcal{T}_k$ is the set of time periods within TOU period k . As is done in [20], we also discourage biogas tank depletion by penalizing deviations in biogas storage between the beginning and end of the day. We define the biogas volume deviation as v_{diff}

$$v_{diff} \geq v^{|\mathcal{T}|} - v^0, \quad (18)$$

$$v_{diff} \geq v^0 - v^{|\mathcal{T}|}, \quad (19)$$

where v^0 and $v^{|\mathcal{T}|}$ are the initial and final tank volumes over the scheduling horizon. This constraint is included to prevent the tank from being fully depleted over the day, which could limit the operational flexibility

within the next day-ahead scheduling period. The total scheduling cost is then

$$F_1(\mathbf{x}) = \sum_{(h,i) \in \mathcal{T}} (\pi_E^h \cdot p_g^{h,i} \cdot \Delta T_{RT} + \pi_b \cdot b_{out}^{h,i} + \alpha_1 \cdot q^{h,i}) + \sum_{k \in \text{TOU}} \pi_D^k \cdot p_D^k + \alpha_2 \cdot v_{diff}, \quad (20)$$

where π_E^h is the price of electricity, π_b is the price of processing biogas, π_D^k is the demand charge at TOU period k , and parameters α_1 and α_2 are penalty factors.

2.3.2. Regulation Revenue. The revenue from providing frequency regulation is based on the day-ahead and real-time committed capacity

$$F_2(\mathbf{x}) = \sum_{h \in \mathcal{T}_h} \pi_{DA,up}^h \cdot R_{DA,up}^h + \pi_{DA,dn}^h \cdot R_{DA,dn}^h, \quad (21)$$

$$F_3(\mathbf{x}) = \sum_{(h,i) \in \mathcal{T}} \pi_{RT,up}^{h,i} \cdot R_{RT,up}^{h,i} + \pi_{RT,dn}^{h,i} \cdot R_{RT,dn}^{h,i}, \quad (22)$$

where $\pi_{DA,up}^h$ and $\pi_{DA,dn}^h$ are the day-ahead regulation up and down prices for hour h , and $\pi_{RT,up}^{h,i}$ and $\pi_{RT,dn}^{h,i}$ are the real-time regulation up and down prices for interval (h, i) .

2.4. Deterministic Formulation

Here, we first write out the deterministic version of the problem, where we assume that we know the regulation signal and biogas production in advance

$$\begin{aligned} \min_{\mathbf{x}} \quad & F_1(\mathbf{x}) - F_2(\mathbf{x}) - F_3(\mathbf{x}) \quad \textbf{(DART-Det)} \\ \text{s.t.} \quad & (1), (2), (4) - (7), (9) - (19), \end{aligned}$$

where decision variable $\mathbf{x}$ is

$$\mathbf{x} = \begin{cases} v_{diff} & \\ R_{DA,up}^h, R_{DA,dn}^h, z^h & \forall h \in \mathcal{T}_h, \\ R_{RT,up}^{h,i}, R_{RT,dn}^{h,i} & \forall (h, i) \in \mathcal{T}, \\ q^{h,i}, v^{h,i}, b_{out}^{h,i}, p_g^{h,i}, p_b^{h,i} & \forall (h, i) \in \mathcal{T}, \\ p_D^k & \forall k \in \text{TOU}. \end{cases}$$

The deterministic problem is a mixed-integer linear program.

3. Multi-Stage Robust Framework

In practice, we do not know the regulation signals or biogas production in advance. While the ACS is constrained to lie between -1 and 1 , the signal

is determined in real time based on fluctuations in system frequency. Additionally, biogas production is a by-product of the anaerobic digestion treatment process, which depends on the uncertain and time-varying quantity and composition of the incoming sewage sludge.

Incorporating these uncertainties into the deterministic problem results in a multi-stage robust problem. In this setting, the day-ahead regulation capacity is determined before the uncertainty is realized, while the real-time regulation capacity decisions are made over the day as the biogas production and regulation signal become known.

In this section, we define the uncertainty set, describe how the uncertainty is incorporated into the optimization problem, and present a tractable reformulation using affine control policies to get an affinely adjustable robust counterpart.

3.1. Incorporating Uncertainty

Here, we consider the biogas production and regulation signals $\tilde{s}^{h,i,\ell}$ to be uncertain but bounded. Specifically, we re-write the biogas production as the sum of the forecast and the uncertain forecast error, i.e.,

$$b_{in}^{h,i} = \hat{b}_{in}^{h,i} + \tilde{b}_{in}^{h,i} \quad \forall (h, i) \in \mathcal{T}, \quad (23)$$

where $\hat{b}_{in}^{h,i}$ is the biogas production forecast and $\tilde{b}_{in}^{h,i}$ is the uncertain-but-bounded biogas production forecast error. In our case study, we define the biogas production forecast error to be bounded by a percentage σ of the biogas production forecast, i.e., $\tilde{b}_{in}^{h,i}$ is in $[-\sigma \hat{b}_{in}^{h,i}, \sigma \hat{b}_{in}^{h,i}]$.

For regulation, the participating generators and loads bid their available regulation capacity, but the actual real-time signal they must respond to is unknown at the time of commitment. The ACS signal lies between -1 and 1 , where the real-time adjustments are scaled by the committed regulation capacity and must remain feasible for any dispatch signal.

In our formulation, the smallest time step for decisions are made at a 15-minute resolution for real-time regulation capacity. To align the four-second regulation signal with this decision resolution, we average the signal over each 15-minute interval

$$\tilde{s}^{h,i} \triangleq \frac{\Delta T_{reg}}{\Delta T_{RT}} \sum_{\ell} \tilde{s}^{h,i,\ell} \quad \forall (h, i) \in \mathcal{T}.$$

We can then write the uncertainty as

$$\mathbf{u}^{h,i} = \begin{bmatrix} \tilde{s}^{h,i} \\ \tilde{b}_{in}^{h,i} \end{bmatrix} \in \mathcal{U}^{h,i}, \quad (24)$$

where the full uncertainty set over the entire scheduling horizon can be written as $\mathcal{U} = \prod_{(h,i) \in \mathcal{T}} \mathcal{U}^{h,i}$.

We assume that the WWTP is a price taker in the ancillary services market. That is, its bids are unable to influence the market clearing prices for regulation. This allows us to use the regulation price forecasts within our formulation. While actual prices are set via market clearing and may differ from the forecasts, we assume that the regulation price is accurate in our formulation. Future work may include price uncertainty or bidding strategies within the robust formulation.

3.2. Multi-Stage Robust Problem

After including the uncertainty set $\mathcal{U}$, the deterministic problem (DART-Det) can then be written as a multi-stage robust problem, where the day-ahead regulation capacity is solved the day before and the real-time regulation capacity is solved at least 75 minutes before the market closes for the trading hour. The general form of multistage robust problem is

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{y}(\cdot)} \quad & G_1(\mathbf{x}) + \max_{\mathbf{u} \in \mathcal{U}} G_2(\mathbf{x}, \mathbf{y}(\mathbf{u})) \\ \text{s.t.} \quad & A(\mathbf{u})\mathbf{x} + B\mathbf{y}(\mathbf{u}) \leq d \quad \forall \mathbf{u} \in \mathcal{U}, \end{aligned}$$

where $G_1(\cdot)$, $G_2(\cdot)$, $A(\cdot)$, and $B(\cdot)$ are functions, $\mathbf{x}$ is the here-and-now decision variables, and $\mathbf{y}$ is the wait-and-see decision variables. The here-and-now decision variables are first-stage decision variables that are made before the uncertain parameters are known. In our problem, this is the scheduled biogas generator operation as well as the day-ahead regulation up and down capacity. The wait-and-see decision variables are the later-stage decisions that are made after the uncertainty is partially or fully realized. In our problem, this is the real-time regulation up and down capacity given the now-known regulation signals and biogas forecast error from earlier in the day. The uncertainty $\mathbf{u} \in \mathcal{U}$ is the range of possible uncertainty realizations where we assume that the uncertainty lies within a convex, bounded set. Specifically, we consider bounding boxes, but other convex sets can lead to other tractable formulations (e.g., ellipsoids). Our uncertainty set $\mathcal{U}$ contains the biogas production forecast error and regulation signal. It should be noted that the multistage robust formulation can also be written in a nested form for the N stages, see e.g., [26].

In our problem, uncertainty is progressively revealed over the day, and real-time regulation capacity bids must be submitted at least 75 minutes before the trading hour. This time-delayed and rolling decision structure makes the problem inherently multi-stage (rather than two-stage). Plugging in the uncertainty definitions (23)-(33) into the deterministic problem (DART-Det),

we get our initial multi-stage robust problem, where (6) and (13) must hold for all uncertainty realizations $\mathbf{u} \in \mathcal{U}$. The here-and-now decision variable $\mathbf{x}$ is

$$\mathbf{x} = \begin{cases} v_{\text{diff}} & \\ R_{\text{DA},\text{up}}^h, R_{\text{DA},\text{dn}}^h, z^h & \forall h \in \mathcal{T}_h, \\ q^{h,i}, v^{h,i}, b_{\text{out}}^{h,i}, p_g^{h,i}, p_b^{h,i} & \forall (h,i) \in \mathcal{T}, \\ p_D^k & \forall k \in \text{TOU}, \end{cases}$$

and the wait-and-see decision variable $\mathbf{y}$ is

$$\mathbf{y} = \left\{ R_{\text{RT},\text{up}}^{h,i}, R_{\text{RT},\text{dn}}^{h,i} \quad \forall (h,i) \in \mathcal{T}. \right.$$

3.3. Real-Time Regulation Control Policies

In their general form, multi-stage robust problems are computationally intractable; however, a simplified affine control policy (or recourse function) can be used to efficiently approximate the wait-and-see decisions as affine functions of previously observed uncertainty realizations [27]. This approach enables tractability while preserving adaptability to real-time conditions. Following the approach in [20], we focus on symmetric regulation, where regulation up and regulation down are equal. We denote the day-ahead and real-time regulation capacities as R_{DA}^h and $R_{\text{RT}}^{h,i}$, which are defined in (14) and (15), respectively.

The real-time regulation capacity is determined based on partially revealed uncertainty. Within CAISO, the regulation capacity for interval (h,i) must be determined at least 75 minutes in advance. This is time delay is illustrated in Fig. 1. As a result, the real-time regulation capacity at time (h,i) is based on uncertainty observed up to time $(h,i) - \delta t$, where $\delta t = 9$ intervals. The affine control policy is then defined as

$$R_{\text{RT}}^{h,i} = B_0^{h,i} + B_{\text{RT}}^{h,i} \cdot \mathbf{u}^{(h,i)-\delta t} \quad (25)$$

where $B_0^{h,i} \in \mathbb{R}$ and $B_{\text{RT}}^{h,i} \in \mathbb{R}^{(1,2)}$ are first-stage decision variables we solve for in the day-ahead problem.

3.4. Affinely Adjustable Robust Counterpart

We formulate and solve the multi-stage robust problem as an affinely adjustable robust counterpart by plugging in the affine control policy (25) into constraints (10)-(13)

$$p_b^{h,i} + R_{\text{DA}}^h + B_0^{h,i} + B_{\text{RT}}^{h,i} \mathbf{u}^{(h,i)-\delta t} \leq \overline{p}_b \cdot z^h, \quad (26)$$

$$p_b^{h,i} - R_{\text{DA}}^h - B_0^{h,i} - B_{\text{RT}}^{h,i} \mathbf{u}^{(h,i)-\delta t} \geq \underline{p}_b \cdot z^h, \quad (27)$$

$$0 \leq v^{h,i} + u_2^{h,i} - \frac{\Delta T_{RT}}{\psi} \cdot \left(R_{DA}^h u_1^{h,i} + B_0^{h,i} + B_{RT}^{h,i} u^{(h,i)-\delta t} \right) \leq \bar{v} + q^{h,i}, \quad (28)$$

$$B_0^{h,i} + B_{RT}^{h,i} u^{(h,i)-\delta t} \geq 0, \quad (29)$$

for all $u \in \mathcal{U}$, $(h, i) \in \mathcal{T}$. The adjusted day-ahead and real-time regulation cost functions become

$$F_2'(x) = \sum_{h \in \mathcal{T}_h} (\pi_{DA,up}^h + \pi_{DA,dn}^h) \cdot R_{DA}^h, \quad (30)$$

$$F_3'(x) = \sum_{(h,i) \in \mathcal{T}} \left((\pi_{RT,up}^{h,i} + \pi_{RT,dn}^{h,i}) \cdot B_0^{h,i} - (\pi_{RT,up}^{h,i} + \pi_{RT,dn}^{h,i}) \cdot B_{RT}^{h,i} u^{(h,i)-\delta t} \right). \quad (31)$$

The coefficients of the affine control policy— $B_0^{h,i}$ and $B_{RT}^{h,i}$ —are first-stage decision variables. The full robust problem can then be written as

$$\min_x F_1(x) - F_2'(x) - F_3'(x) \quad (\mathbf{DART-Robust})$$

$$\text{s.t.} \quad (1), (2), (4) - (5), (7), (16) - (19), \\ (26) - (29),$$

$$v^{h,i} = v^{(h,i)-1} + \hat{b}_{in}^{h,i} - b_{out}^{h,i} \quad \forall (h, i) \in \mathcal{T},$$

$$R_{DA}^h \geq 0 \quad \forall h \in \mathcal{T}_h,$$

where the here-and-now decision variable x is

$$x = \begin{cases} v_{diff} & \forall h \in \mathcal{T}_h, \\ R_{DA}^h, z^h, & \forall h \in \mathcal{T}_h, \\ B_0^{h,i}, B_{RT}^{h,i} & \forall (h, i) \in \mathcal{T}, \\ q^{h,i}, v^{h,i}, b_{out}^{h,i}, p_g^{h,i}, p_b^{h,i} & \forall (h, i) \in \mathcal{T}, \\ p_D^k & \forall k \in \text{TOU}. \end{cases}$$

We reformulate the uncertain constraints using the explicit maximization approach in [28]. Specifically, when the constraints are linear with respect to the decision variables and uncertainty, i.e.,

$$c^\top u + d + (Au)^\top x + b^\top x \leq 0 \quad \forall u \in \{u : \|u\|_\infty \leq 1\},$$

we can rewrite this as

$$(b^\top x + d) + \|c + A^\top x\|_1 \leq 0,$$

where $\|\cdot\|_\infty$ and $\|\cdot\|_1$ are an L_∞ -norm and an L_1 -norm. For instance, (26) can be equivalently reformulated as

$$p_b^{h,i} + R_{DA}^h + B_0^{h,i} + \|D^{(h,i)-\delta t} B_{RT}^{h,i}\|_1 \leq \bar{p}_b \cdot z^h, \quad (32)$$

Table 1. PG&E summer electric rates [29]

TOU	k	Demand Charge π_D^k (\$/kW)	Energy Charge π_E^h (\$/kWh)
Peak	1	48.55	0.20
Partial Peak	2	10.33	0.16
Off-Peak	3	—	0.13
Overall	4	36.95	—

for all $(h, i) \in \mathcal{T}$, where $D^{h,i}$ is a scaling matrix

$$D^{h,i} = \begin{bmatrix} 1 & 0 \\ 0 & \sigma \hat{b}_{in}^{h,i} \end{bmatrix}. \quad (33)$$

The L1-norm term can be modeled using auxiliary variables and two additional linear constraints. The final formulation remains a MILP, which can be efficiently solved using commercial solvers.

4. Case Study

We evaluate our proposed multi-stage robust approach (**DART-Robust**) that solves for both the day-ahead and real-time regulation capacity and compare it against two benchmarks:

1. a single-stage robust formulation that optimally solves for the day-ahead regulation capacity in advance [20], referred to as (**DA-Robust**), and
2. the deterministic formulation in Section 2.4 where we have perfect knowledge of the uncertainty realizations in advance, (**DART-Det**).

The case study is based on data representative of the San José–Santa Clara Regional Wastewater Facility in California.

4.1. Set Up

The WWTP is located in PG&E's service territory, so we use PG&E's B-20 Industrial/General Service primary electric rate schedule [29]. Table 1 summarizes the energy and demand charges for each TOU period. The peak period is 4-9 pm, the partial-peak periods are 2-4 pm and 9-11 pm, and all other times are off-peak periods. Similar to [20], we do not include the flat daily customer charge as it does not affect the biogas generator dispatch and we normalize the demand charge over the 31 days in July so it is proportional to the energy costs within the 24-hour scheduling horizon.

We obtain day-ahead and real-time regulation up and down prices from CAISO's Open Access Same-Time Information System (OASIS) [30]. We pull the prices for July 2020 and zone NP-26 (i.e., AS_NP26_Exp), so it corresponds with the power consumption information we have for the WWTP. Fig. 3 shows the average

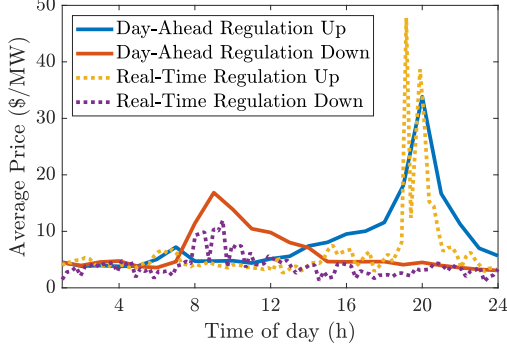


Figure 3. Average CAISO day-ahead and real-time regulation up and down prices over July 2020.

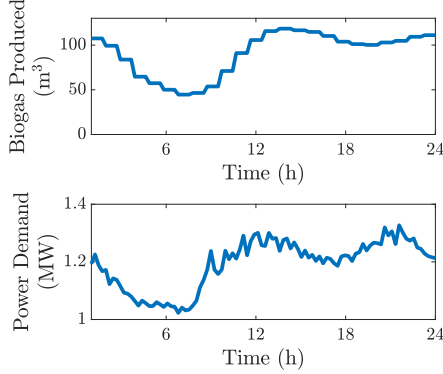


Figure 4. Biogas production forecast (top) and power demand (bottom).

day-ahead and real-time regulation up and down prices over July 2020.

The San José–Santa Clara Regional Wastewater Facility serves around 1.4 million people. It produces biogas through anaerobic digestion and has existing cogenerators and biogas storage tanks. According to facility reports, the plant can offset up to two-thirds of its electricity demand using biogas [31].

Biogas production $\hat{b}_{in}^{h,i}$ is estimated given anaerobic digestion parameters and flow rate patterns as described in [20]. We scale the estimate so biogas can cover two-thirds of the plant’s electricity demand, in line with the treatment plant’s annual reports. The power demand is pulled from [32] for July 2020 at a 15-minute resolution. The power demand $p_{demand}^{h,i}$ and biogas production forecast $\hat{b}_{in}^{h,i}$ are shown in Fig. 4. We set $\pi_b = 0.02 \$/m^3$, $\bar{v} = 1416 m^3$, $\bar{p}_b = 800 kW$ and $\bar{p}_b = 8,000 kW$ [20]. Last, we set the penalty terms α_1 and α_2 to 10,000 and $v_0 = 0.5 \bar{v}$.

To evaluate performance under uncertainty, we generate full-day synthetic regulation signals using the code from [33] and sample biogas production forecast errors from a uniform distribution over the range $[-\sigma \hat{b}_{in}^{h,i}, \sigma \hat{b}_{in}^{h,i}]$ where σ is a fraction.

Table 2. Comparison of Average Costs (\$)

Approach	Schedule	Regulation		Total
		DA	RT	
DART-Det	2,884.11	11.29	232.36	2,640.46
DA-Robust	3,239.81	11.62	—	3,228.19
DART-Robust	3,246.82	2.29	89.14	3,155.39

We compare our approach DART-Robust against a robust approach that only solves for day-ahead regulation capacity DA-Robust and a deterministic approach solving for the day-ahead and real-time regulation capacity assuming perfect knowledge DART-Det. The three formulations are mixed-integer linear programs and are solved using the Gurobi solver through the JuMP optimization package in Julia.

4.2. Results

We first compare the solutions of the three formulations: DART-Det, DA-Robust, and DART-Robust. To evaluate the impact of uncertainty, we assess performance across multiple uncertainty realizations. Specifically, we generate 100 samples of regulation signals and biogas production per day over the entire month of July, i.e., 3,100 samples for each approach. Table 2 compares the average daily schedule cost F_1 (i.e., electricity and biogas processing), day-ahead regulation revenue F_2 , real-time regulation revenue F_3 , and total cost for each approach when $\sigma = 1\%$. As expected, DART-Det achieves the lowest cost; this is because we assume we are able to make decisions with perfect knowledge of future regulation signals and biogas production.

In Table 2, we observe that the DART-Robust solution is less expensive than the DA-Robust solution. This improvement is due to DART-Robust’s ability to adjust real-time regulation capacity based on partial uncertainty realizations, whereas DA-Robust commits to providing day-ahead regulation capacity in advance without the ability to adjust. We also observe that DART-Robust and DART-Det provide more real-time regulation than day-ahead regulation. This is likely due to the CAISO price forecasts, where the solver maximizes revenue when day-ahead and real-time prices are high. This aligns with the average price trends in Fig. 3, where the real-time regulation up prices are especially high during early evening. Both robust approaches respond to these price signals when feasible, with the DART-Robust formulation benefiting from its ability to adjust its real-time regulation capacity given biogas and regulation signal realizations. These results also provide motivation to consider asymmetric regulation to further maximize the WWTP’s participation in the day-ahead and real-time regulation

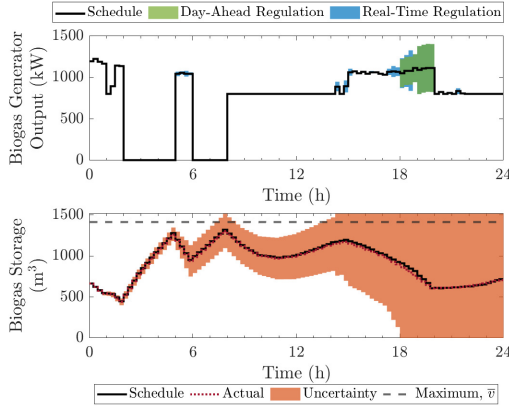


Figure 5. Example DART-Robust schedule (black line) and regulation capacity (green and blue shaded areas). Top: Biogas generator output $p_b^{h,i}$. Bottom: Biogas volume $v^{h,i}$, where the red dotted line is the actual biogas volume and the red shaded region is all possible volume trajectories under uncertainty set $\mathcal{U}$.

markets.

To further illustrate the DART-Robust solution, Fig. 5 shows the biogas generator output and storage volume for one uncertainty realization on July 31st with $\sigma = 5\%$. The top plot shows the biogas generator output, where the green and blue shaded bands around the schedule represent the day-ahead and real-time regulation capacities, respectively. The bottom plot shows the biogas tank volume over the day, where the red dashed line shows the realized trajectory. The red shaded region represents the full range of possible volume trajectories given the uncertainty set. It is important to note that the shaded region represents the full range of volume trajectories *before* any uncertainty is revealed.

This example highlights two insights. First, control policies allows us to provide real-time regulation capacity that would be infeasible under a purely day-ahead strategy. The DART-Robust solution does not hold for all possible uncertainty realizations, where we see in the bottom plot of Fig. 5 that, under certain uncertainty realizations, the plant would need to flare biogas as early as $h = 8$ and would violate the minimum tank level by $h = 18$. However, once the uncertainty starts getting progressively revealed throughout the day, we will see this region shrink, enabling the ability to provide more real-time regulation while ensuring that there are no constraint violations. Fig. 6 illustrates the actual dispatch in a close-up view of a three-hour window where the WWTP is providing both day-ahead and real-time regulation.

Second, the optimal solution does not provide regulation for all time periods. This is because there is a trade-off in regulation revenue against energy cost

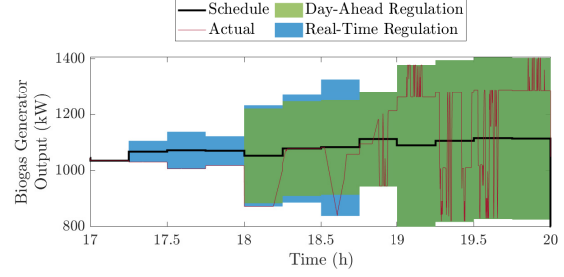


Figure 6. Close-up of biogas generator tracking the regulation signal (red line) for Fig. 5 example.

reduction, where the optimal solution only provides regulation capacity during high-price periods. This behavior is especially apparent around $h = 18$ where regulation prices peak.

Last, we assess the computational complexity of the three approaches, DART-Det, DA-Robust, and DART-Robust. All three formulations are mixed-integer linear programs, which is a class of problem that can be solved using efficient commercial solvers. However, the DART-Robust formulation introduces additional decision variables and constraints in its reformulation to represent the L1-norm term. As a result, the increased size will make it computationally more expensive. The average solver time for the 3,100 samples for DART-Det, DA-Robust, and DART-Robust in Table 2 were 0.1 seconds, 0.1 seconds, and 10.1 seconds, respectively. Despite the solver time increase, all DART-Robust problem remains sufficiently fast for its intended application of solving for the control policy parameters in the day-ahead problem.

5. Conclusion

In this work, we developed a multi-stage robust optimization framework for scheduling a biogas generator at a wastewater treatment plant participating in day-ahead and real-time frequency regulation markets. The formulation accounts for uncertainty in biogas production and the regulation signal, with the schedule and day-ahead regulation capacity determined in the first stage and real-time regulation capacity decisions made as uncertainty is progressively revealed. To reduce conservativeness, we employ affine control policies that determine real-time regulation capacity given partial uncertainty realizations. A case study demonstrates that this approach enables greater regulation participation and lower operational cost compared to a day-ahead-only strategy, while ensuring feasibility under uncertainty. Although more computationally intensive, the method remains tractable for day-ahead operation. Future work will explore asymmetric regulation products and price uncertainty.

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